

Linear Regression

MATH 4432 Statistical Machine Learning
Tutorial T03

Estimation, uncertainty and model diagnostics

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The linear regression model

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

n observations. **p** coefficients, including the intercept.

$$X \in \mathbb{R}^{n \times p}, \quad \boldsymbol{\beta} \in \mathbb{R}^p, \quad x_i^T = (1, x_{i1}, \dots, x_{i,p-1})$$

$$E(\boldsymbol{\varepsilon} \mid X) = 0, \quad \text{Var}(\boldsymbol{\varepsilon} \mid X) = \sigma^2 I_n$$

The conditional mean is linear in the coefficients.

Throughout: condition on X , assume $n > p$ and full column rank. Simple regression has $p = 2$.

Ordinary least squares

$$\hat{\boldsymbol{\beta}} = \operatorname{argmin}_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2$$

Differentiate the residual sum of squares:

$$\nabla_{\boldsymbol{\beta}} \text{RSS} = -2X^T(\mathbf{y} - X\boldsymbol{\beta})$$

$$X^T X \hat{\boldsymbol{\beta}} = X^T \mathbf{y}$$

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}$$

Full column rank makes $X^T X$ invertible and the solution unique.

Gaussian errors give the same estimate

$$\boldsymbol{\varepsilon} \mid X \sim \mathcal{N}(0, \sigma^2 I_n)$$

The log-likelihood is

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2$$

For fixed noise variance, maximizing likelihood minimizes RSS.

OLS therefore equals the Gaussian maximum likelihood estimate.

Normality enables the exact t and F results used later. OLS itself does not require normal errors.

Estimating the noise variance

$$\mathbf{e} = \mathbf{y} - X\hat{\boldsymbol{\beta}}, \quad \text{RSS} = \mathbf{e}^T \mathbf{e}$$

Maximum likelihood

$$\hat{\sigma}_{\text{ML}}^2 = \frac{\text{RSS}}{n}$$

Unbiased estimate

$$s^2 = \frac{\text{RSS}}{n - p}$$

Fitting p coefficients uses p degrees of freedom.

$$E(\text{RSS} \mid X) = (n - p)\sigma^2$$

p includes the intercept. For one predictor, the denominator is $n - 2$.

Why coefficient estimates vary

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (X^T X)^{-1} X^T \boldsymbol{\varepsilon}$$

Repeated samples change the errors and therefore the fitted coefficients.

$$E(\hat{\boldsymbol{\beta}} \mid X) = \boldsymbol{\beta}$$

$$\text{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 (X^T X)^{-1}$$

Noise level and predictor geometry both determine uncertainty.

The variance formula is a covariance matrix. Its diagonal contains coefficient variances.

Standard errors, t tests and confidence intervals

$$SE(\hat{\beta}_j) = s \sqrt{[(X^T X)^{-1}]_{jj}}$$

For a specified null hypothesis:

$$H_0 : \beta_j = b_j, \quad T_j = \frac{\hat{\beta}_j - b_j}{SE(\hat{\beta}_j)} \sim t_{n-p}$$

A 95% confidence interval for one coefficient:

$$\hat{\beta}_j \pm t_{0.975, n-p} SE(\hat{\beta}_j)$$

A small two-sided p-value is evidence against H_0 under the model. It does not measure effect size.

Example: a coefficient and its uncertainty

Boston data: 506 neighborhoods

medv: median house value. lstat: percentage of households with low socioeconomic status.

$$\widehat{\text{medv}} = 34.554 - 0.950 \text{ lstat}$$

Term	Estimate	Standard error	95% interval
Intercept	34.554	0.563	[33.448, 35.659]
lstat	-0.950	0.039	[-1.026, -0.874]

One percentage point more lstat is associated with 0.950 fewer units of fitted medv.

This describes an association in the fitted model, not a causal effect.

Multiple regression changes the interpretation

$$E(Y \mid x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_{p-1} x_{p-1}$$

Each slope describes a change in the conditional mean while holding the other included predictors fixed.

Boston model	Istat coefficient	Training R ²
Istat only	-0.950	0.544
All 12 predictors	-0.552	0.734

Adding predictors changes the adjustment set. A coefficient may change size, sign or significance.

A mean response and a new observation

At a fixed new predictor vector x_0 , including the intercept:

Conditional mean

$$\mu_0 = x_0^T \boldsymbol{\beta}$$

Average response at x_0 .

Uncertainty comes from estimating the coefficients.

New observation

$$Y_0 = x_0^T \boldsymbol{\beta} + \varepsilon_0$$

One future response at x_0 .

Uncertainty also includes its new random error.

$$\hat{\mu}_0 = x_0^T \hat{\boldsymbol{\beta}}$$

Confidence interval for the mean response

$$h_0 = x_0^T (X^T X)^{-1} x_0$$

The fitted mean inherits uncertainty from the estimated coefficients:

$$\text{Var}(\hat{\mu}_0 \mid X) = \sigma^2 h_0$$

A pointwise 95% confidence interval for the true mean μ_0 :

$$\hat{\mu}_0 \pm t_{0.975, n-p} s \sqrt{h_0}$$

Across repeated training samples, 95% of intervals constructed this way contain the true mean at x_0 .

Prediction interval for a new observation

$$Y_0 - \hat{\mu}_0 = \varepsilon_0 - x_0^T(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$$

Assume the new error is independent of the training errors.

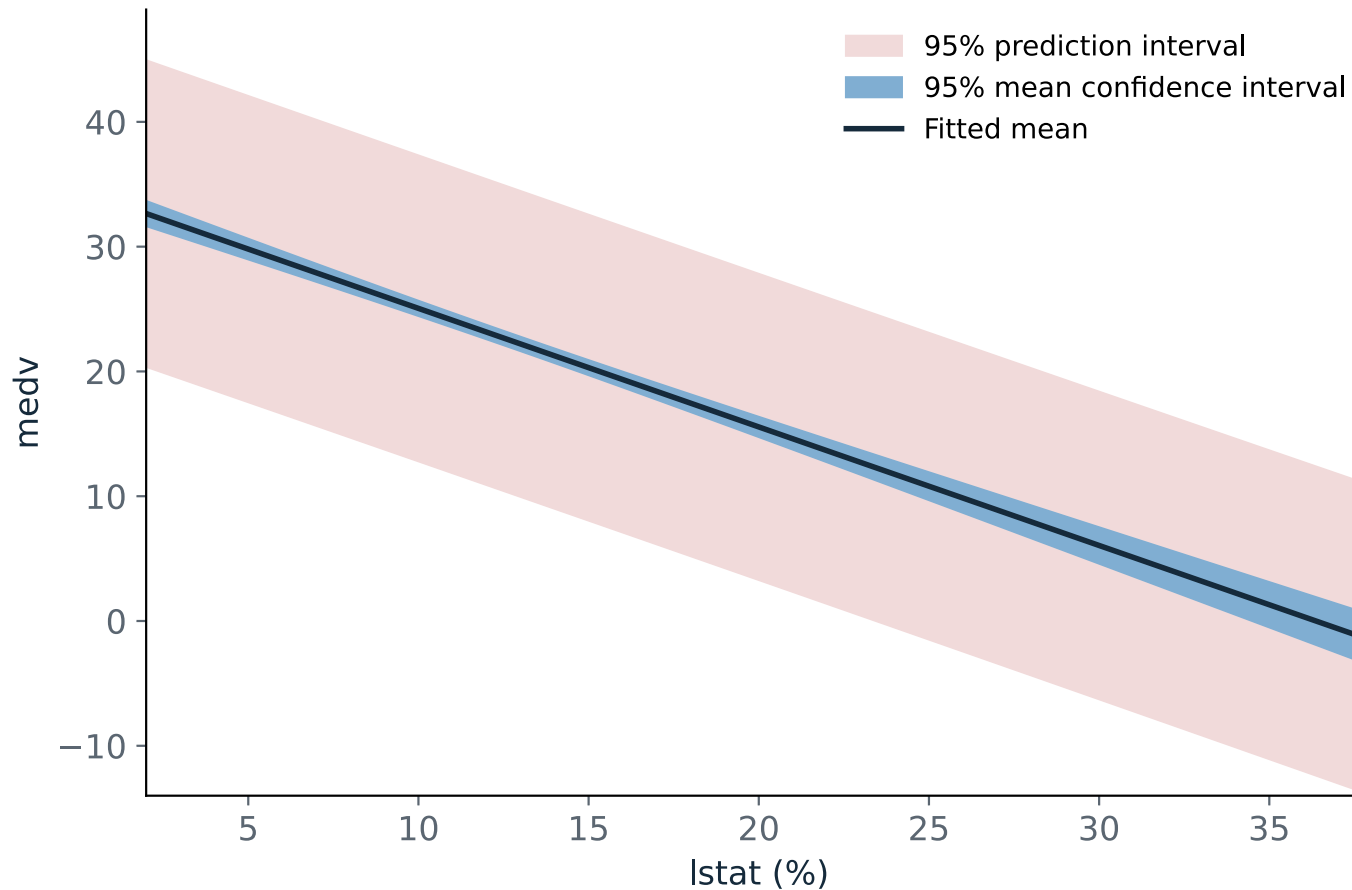
$$\text{Var}(Y_0 - \hat{\mu}_0 \mid X) = \sigma^2 + \sigma^2 h_0$$

A pointwise 95% prediction interval for Y_0 :

$$\hat{\mu}_0 \pm t_{0.975, n-p} s \sqrt{1 + h_0}$$

The extra 1 represents fresh observation noise. At the same x_0 and confidence level, the PI is wider when $s > 0$.

The extra noise makes the prediction interval wider



At Istat = 10%

Fitted medv: **25.05**

95% mean CI
[24.47, 25.63]

95% prediction interval
[12.83, 37.28]

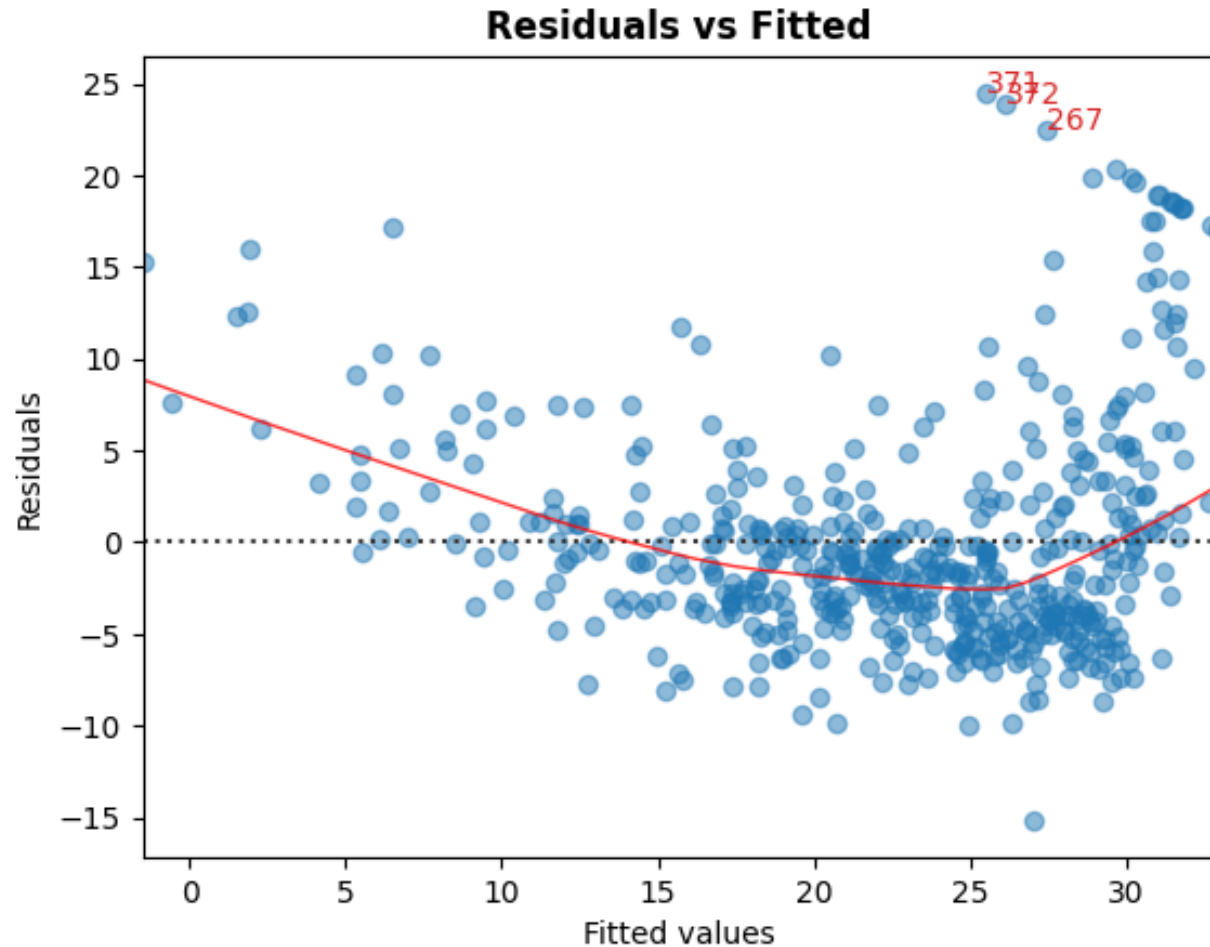
These are model-based, pointwise intervals. Their coverage depends on the regression assumptions.

Diagnostics check the assumptions behind inference

Assumption	What to examine	Warning sign
Linear mean	Residuals vs fitted	Curved pattern
Independent errors	Residuals vs time/order	Runs or serial pattern
Normal errors	Normal Q-Q plot	Systematic tail departures
Equal variance	Scale-location plot	Changing vertical spread

Also examine unusual observations and collinearity. A large R^2 does not validate these assumptions.

Residuals reveal a missing pattern in the mean

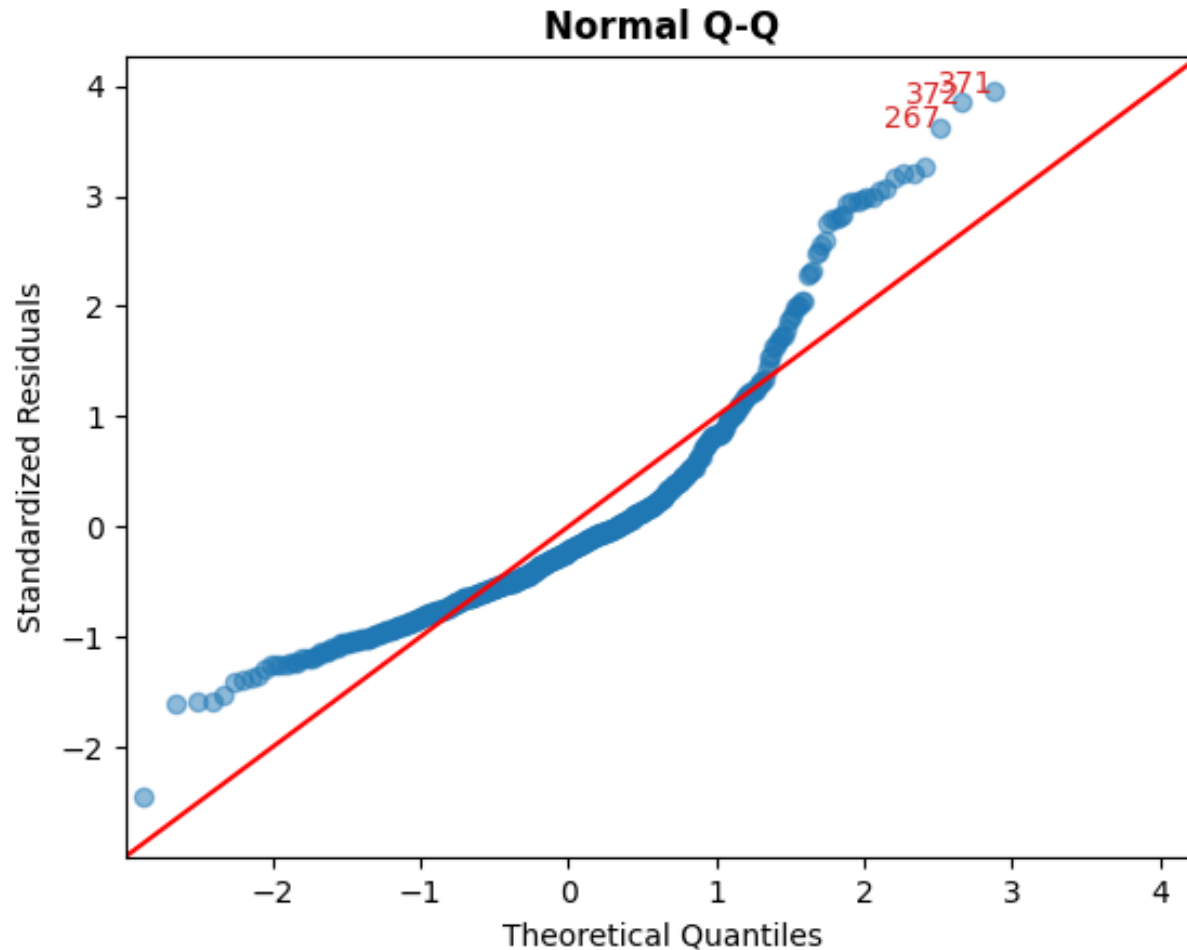


Residuals should fluctuate around zero without a systematic curve.

Here the smooth curve bends, suggesting that a straight line misses structure.

Boston: medv regressed on lstat. A residual is an observed value minus its fitted value.

A Q-Q plot checks the shape of the errors

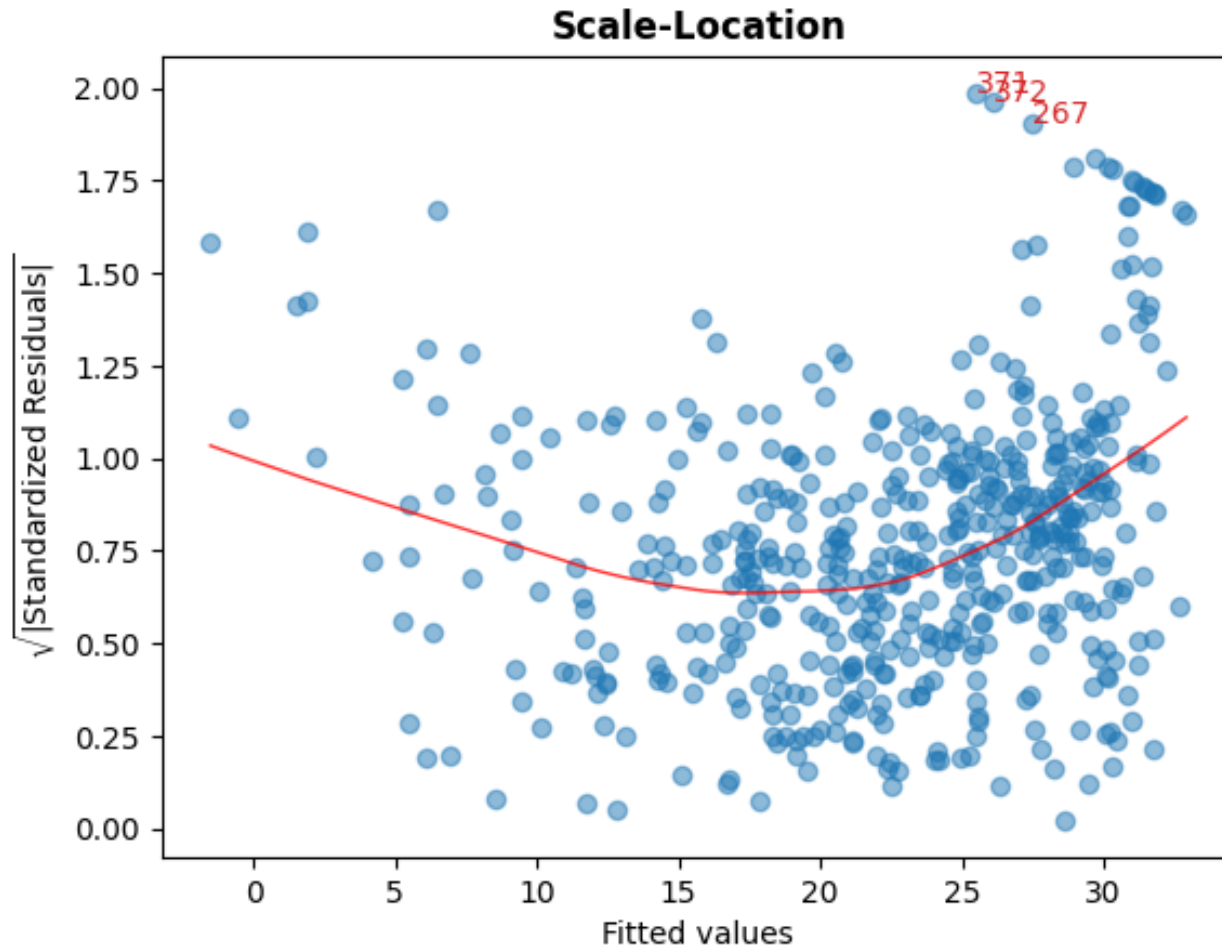


Approximately normal errors give residual quantiles close to a straight line.

The large upper-tail departures here question the normal-error approximation.

The normality assumption concerns errors conditional on predictors, not the marginal response.

Scale-location checks whether the spread is stable



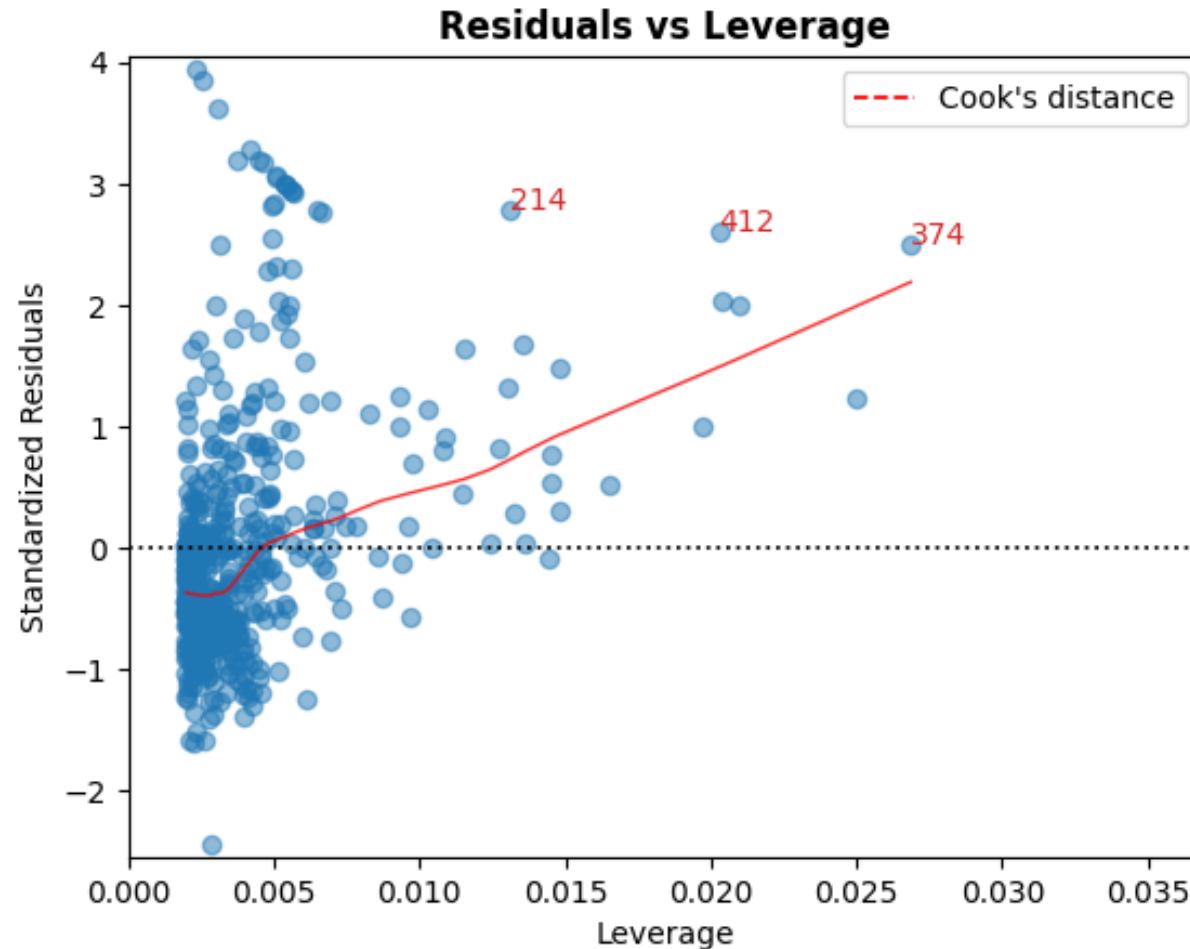
The vertical axis is the square root of the absolute standardized residual.

A roughly flat trend supports constant variance.

Changing spread may call for a better mean model or a transformation.

Recheck the spread after improving the mean. Curvature can also distort this diagnostic.

Outliers, leverage and influence



Outlier

An unusually large standardized residual.

High leverage

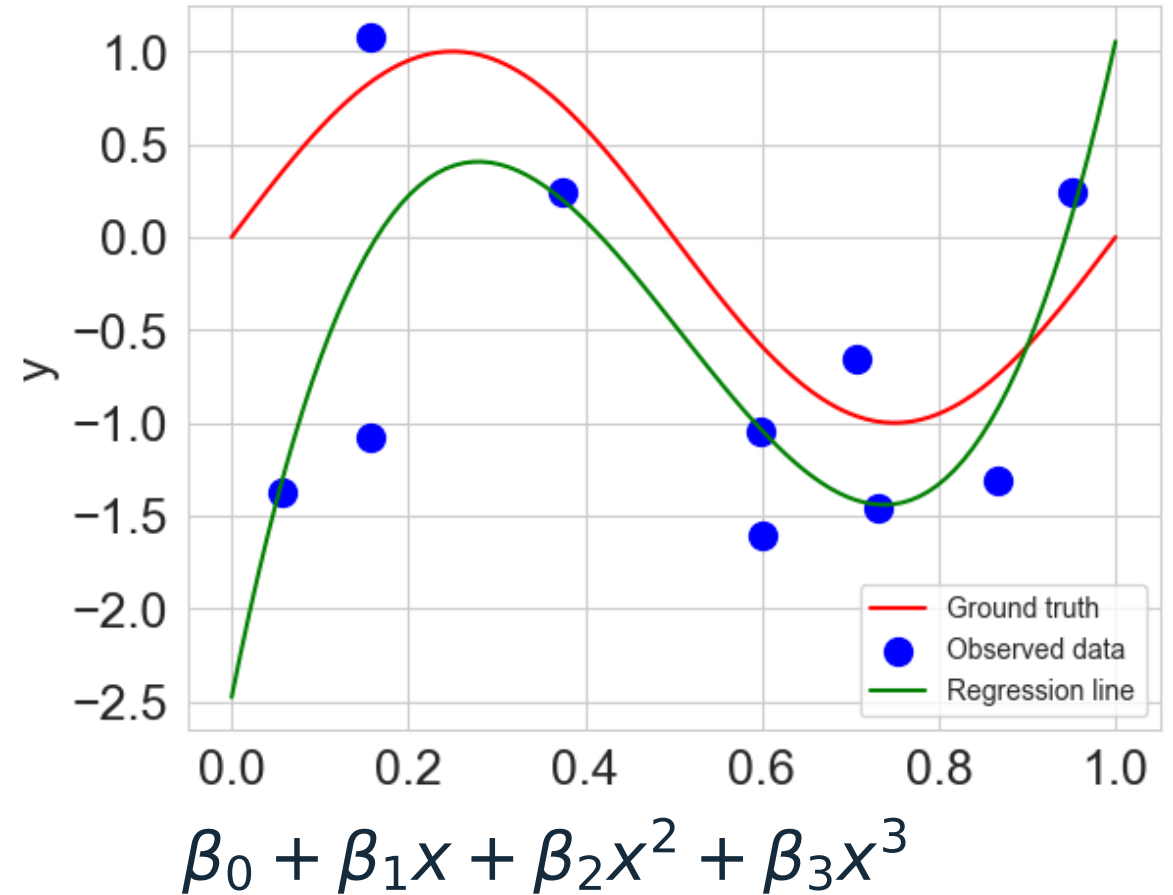
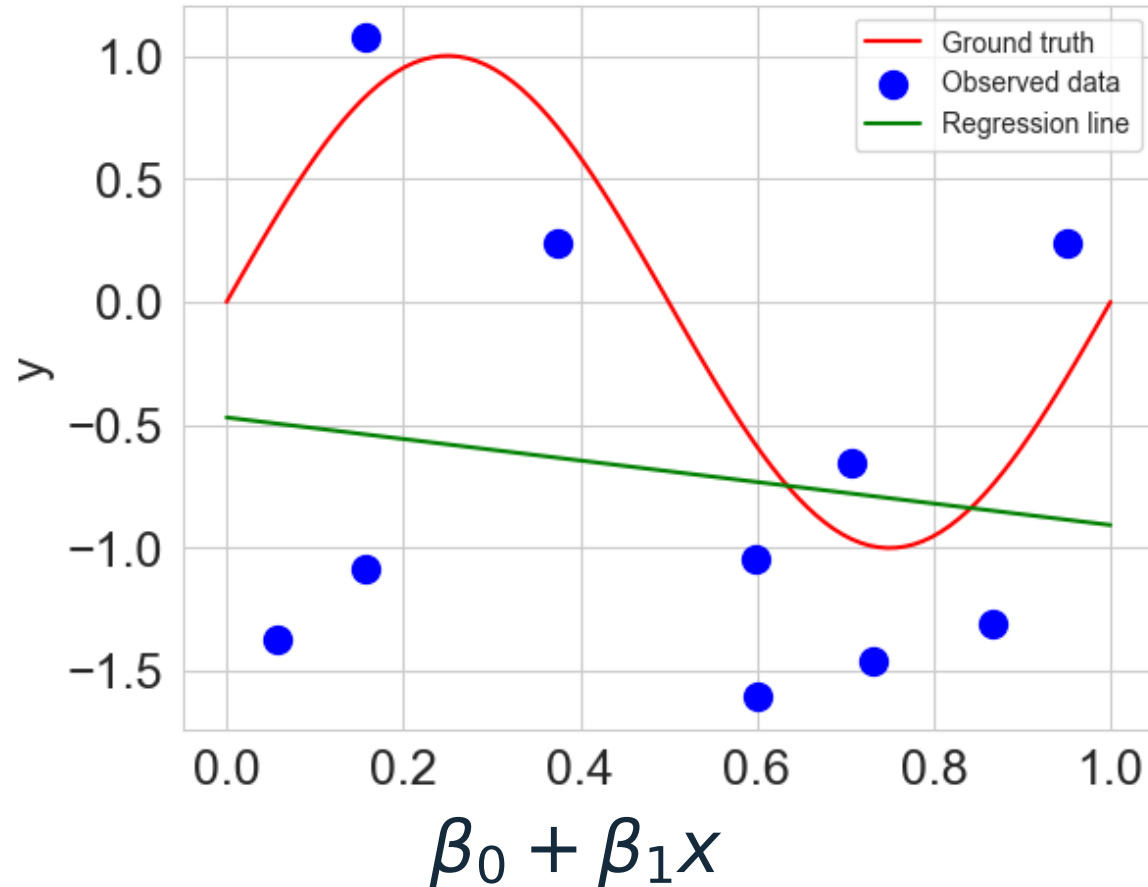
An unusual predictor vector relative to the sample.

Influence

How much deleting a point changes the fit. Cook's distance measures this.

Leverage alone does not imply influence. Investigate unusual points before deciding how to handle them.

Polynomial features allow a curved fitted mean



The same 10 observations. Both fits are linear in their coefficients, even when the curve bends.

An interaction lets one slope depend on another feature

$$E(Y \mid x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

$$\frac{\partial E(Y \mid x_1, x_2)}{\partial x_1} = \beta_1 + \beta_3 x_2$$

Boston example: lstat, age and their interaction

$$\text{slope for lstat} = -1.3921 + 0.0042 \text{ age}$$

The lstat association changes with age. The lstat main coefficient refers to age = 0.

Centering age makes the main coefficient refer to a more useful reference value.

Transforming a predictor changes the mean function

Polynomial predictor

$$\mu(x) = \beta_0 + \beta_1 x + \beta_2 x^2$$

The local slope changes with x .

$$\mu'(x) = \beta_1 + 2\beta_2 x$$

Log predictor

$$\mu(x) = \beta_0 + \beta_1 \log x$$

Requires $x > 0$. A proportional change in x gives an additive change in the mean.

$$\mu(cx) - \mu(x) = \beta_1 \log c$$

Examples: lstat^2 and $\log(\text{rm})$, with rm = average rooms per house. Both keep the original response scale.

A better training fit still needs checking

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}, \quad \text{TSS} = \sum_i (y_i - \bar{y})^2$$

Boston specification

Training R^2

lstat

0.544

lstat and lstat²

0.641

All 12 predictors

0.734

Adding terms to a nested OLS model cannot increase training RSS.

Training R^2 alone cannot establish model validity or predictive accuracy.

Practice: interpreting an Auto regression

392 cars. Response: mpg. Predictor: horsepower.

$$\widehat{\text{mpg}} = 39.9359 - 0.1578 \text{ horsepower}$$

1. Interpret the slope in words.
2. What is the fitted mpg at horsepower = 98?
3. Which interval describes the mean mpg, and which describes one new car?

The slope has a 95% confidence interval of [-0.171, -0.145] in the simple model.

Practice: interpretation and adjustment

Simple regression

Each additional unit of horsepower is associated with about 0.158 fewer mpg in the fitted mean.

$$\widehat{\text{mpg}}(98) = 39.9359 - 0.1578(98) \approx 24.47$$

The mean CI targets the average mpg at 98 horsepower.

The wider PI targets the mpg of one new car.

Multiple regression: the fitted year coefficient is 0.7508.

Holding the other included predictors fixed, one later model year is associated with about 0.751 more mpg.

Key ideas

Estimation chooses coefficients that minimize RSS. Gaussian likelihood gives the same coefficients.

Inference uses noise variance and the geometry of X to quantify coefficient uncertainty.

Prediction distinguishes the uncertainty in a mean from the additional noise in a new observation.

Diagnostics guide changes to the model. Polynomial and interaction terms still fit within linear regression.

Appendix: residuals and degrees of freedom

$$H = X(X^T X)^{-1} X^T, \quad \hat{\mathbf{y}} = H\mathbf{y}, \quad \mathbf{e} = (I - H)\mathbf{y}$$

$$\text{Var}(\mathbf{e} \mid X) = \sigma^2(I - H)$$

$$\text{Var}(e_i \mid X) = \sigma^2(1 - h_{ii}), \quad r_i = \frac{e_i}{s\sqrt{1 - h_{ii}}}$$

Residuals generally depend on one another, even when the errors are independent.

$$E(\text{RSS} \mid X) = \sigma^2 \text{tr}(I - H) = \sigma^2(n - p)$$

Here r_i is an internally standardized residual. H is an orthogonal projector with rank p .

Appendix: information and collinearity

$$-\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^T} = \frac{X^T X}{\sigma^2}, \quad \mathcal{I}_{\boldsymbol{\beta}\boldsymbol{\beta}}^{-1} = \sigma^2 (X^T X)^{-1}$$

Nearly redundant predictors make coefficient estimates unstable.

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

R_j^2 comes from regressing predictor j on the other predictors, with an intercept.

A large VIF flags inflated coefficient variance. Exact redundancy violates the full-rank assumption.

Appendix: testing all slopes together

$$H_0 : \beta_1 = \cdots = \beta_{p-1} = 0$$

Compare the full model with an intercept-only model.

$$F = \frac{(\text{TSS} - \text{RSS})/(p-1)}{\text{RSS}/(n-p)} \sim F_{p-1, n-p} \quad \text{under } H_0$$

A significant F test provides evidence that at least one slope is nonzero. It does not identify which one.